Central Limit Theorem Simulation with Exponential Distribution

Objective

Demonstrate the Central Limit Theorem by examining how the sampling distribution of means approaches normality as sample size increases, even when sampling from a highly skewed exponential distribution.

Problem Statement

Part (a): Generate 1500 random samples of size n = 10 from an exponential distribution with mean = 2 and standard deviation = 2. Calculate the sample mean for each sample and draw a relative frequency histogram.

Part (b): Repeat part (a) with n = 50 and n = 100, and draw conclusions about the behavior of the sampling distribution.

```
# Install packages if needed (uncomment if running for the first time)
install.packages("ggplot2")
install.packages("gridExtra")
install.packages("dplyr")

# Load required libraries
library(ggplot2)
library(gridExtra)
library(dplyr)

cat("✓ All libraries loaded successfully!\n")
```

```
Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

✓ All libraries loaded successfully!
```

Simulation Parameters

Population Distribution

- **Distribution:** Exponential
- **Mean (μ):** 2
- **Standard Deviation (σ):** 2
- **Rate parameter (λ):** 1/2 = 0.5

Simulation Settings

- **Number of samples:** 1,500 for each sample size
- **Sample sizes to test:** n = 10, 50, 100

Key Theoretical Prediction

According to the Central Limit Theorem, the sampling distribution of means should have:

- Mean = μ = 2
- Standard Error = σ/√n

```
set.seed(42)

#P parameters
POPULATION_MEAN <- 2
POPULATION_SD <- 2

#rate params for e distribution
#exponential: mean = 1/lambda, so lambda = 1/mean
RATE_PARAMETER <- 1 / POPULATION_MEAN

# Simulation params
N_SAMPLES <- 1500
SAMPLE_SIZES <- c(10, 50, 100)

cat("Configuration:\n")
cat(sprintf("  Population Mean: %.2f\n", POPULATION_MEAN))
cat(sprintf("  Population SD: %.2f\n", POPULATION_SD))
cat(sprintf("  Rate Parameter (λ): %.2f\n", RATE_PARAMETER))
cat(sprintf("  Number of Samples: %d\n", N_SAMPLES))
cat(sprintf("  Sample Sizes: %s\n", paste(SAMPLE_SIZES, collapse = ", ")))
```

```
Configuration:
  Population Mean: 2.00
  Population SD: 2.00
  Rate Parameter (λ): 0.50
  Number of Samples: 1500
  Sample Sizes: 10, 50, 100
```

Core Functions

We'll define three main functions:

1. `generate_sample_means()` - Efficiently generates multiple samples and calculates their means using matrix operations
2. `calculate_summary_stats()` - Computes descriptive statistics for the sampling distribution
3. `create_histogram()` - Creates professional visualizations with:
   - Relative frequency histogram
   - Theoretical normal distribution overlay
   - Population mean reference line

```
#' Generate Sample Means from Exponential Distribution
#'
```

```
#
#' @param n_samples Integer. Number of samples to generate
#' @param sample_size Integer. Size of each sample
#' @param rate Numeric. Rate parameter for exponential distribution
#' @return Numeric vector of sample means
generate_sample_means <- function(n_samples, sample_size, rate) {
  # Generate matrix: each row is a sample
  sample_matrix <- matrix(
    rexp(n_samples * sample_size, rate = rate),
    nrow = n_samples,
    ncol = sample_size
  )

  # Calculate mean of each sample (row)
  sample_means <- rowMeans(sample_matrix)

  return(sample_means)
}

#' Calculate Summary Statistics
#'
#' @param x Numeric vector
#' @return Named list of summary statistics
calculate_summary_stats <- function(x) {
  list(
    mean = mean(x),
    sd = sd(x),
    median = median(x),
    min = min(x),
    max = max(x),
    q25 = quantile(x, 0.25),
    q75 = quantile(x, 0.75)
  )
}

#' Create Professional Histogram with Normal Overlay
#'
#' @param data Numeric vector of sample means
#' @param sample_size Integer. Size of each sample
#' @param population_mean Numeric. True population mean
#' @param population_sd Numeric. True population standard deviation
#' @return ggplot object
create_histogram <- function(data, sample_size, population_mean, population_sd) {
  # Calculate theoretical standard error
  theoretical_se <- population_sd / sqrt(sample_size)

  # Create data frame for plotting
  df <- data.frame(sample_means = data)

  # Calculate histogram breaks
  n_bins <- 30

  # Create the plot
  p <- ggplot(df, aes(x = sample_means)) +
    # Histogram with relative frequency
    geom_histogram(
      aes(y = after_stat(density)),
      bins = n_bins,
      fill = "#4A90E2",
      color = "white",
      alpha = 0.7
    ) +
    # Overlay theoretical normal distribution
```

```
    stat_function(
      fun = dnorm,
      args = list(mean = population_mean, sd = theoretical_se),
      color = "#E74C3C",
      linewidth = 1.2,
      linetype = "solid"
    ) +
    # Add vertical line at population mean
    geom_vline(
      xintercept = population_mean,
      color = "#27AE60",
      linetype = "dashed",
      linewidth = 1
    ) +
    # Labels and theme
    labs(
      title = sprintf("Sampling Distribution of Mean (n = %d)", sample_size),
      subtitle = sprintf(
        "Mean = %.3f, SD = %.3f, Theoretical SE = %.3f",
        mean(data), sd(data), theoretical_se
      ),
      x = "Sample Mean",
      y = "Relative Frequency (Density)",
      caption = sprintf(
        "Red curve: Normal(mu=%.1f, sigma=%.3f) | Green line: Population mean",
        population_mean, theoretical_se
      )
    ) +
    theme_minimal(base_size = 11) +
    theme(
      plot.title = element_text(face = "bold", size = 13),
      plot.subtitle = element_text(color = "gray40", size = 10),
      plot.caption = element_text(color = "gray50", size = 8, hjust = 0),
      panel.grid.minor = element_blank(),
      panel.grid.major = element_line(color = "gray90")
    )

  return(p)
}

cat("✓ All functions defined successfully!\n")
```

```
✓ All functions defined successfully!
```

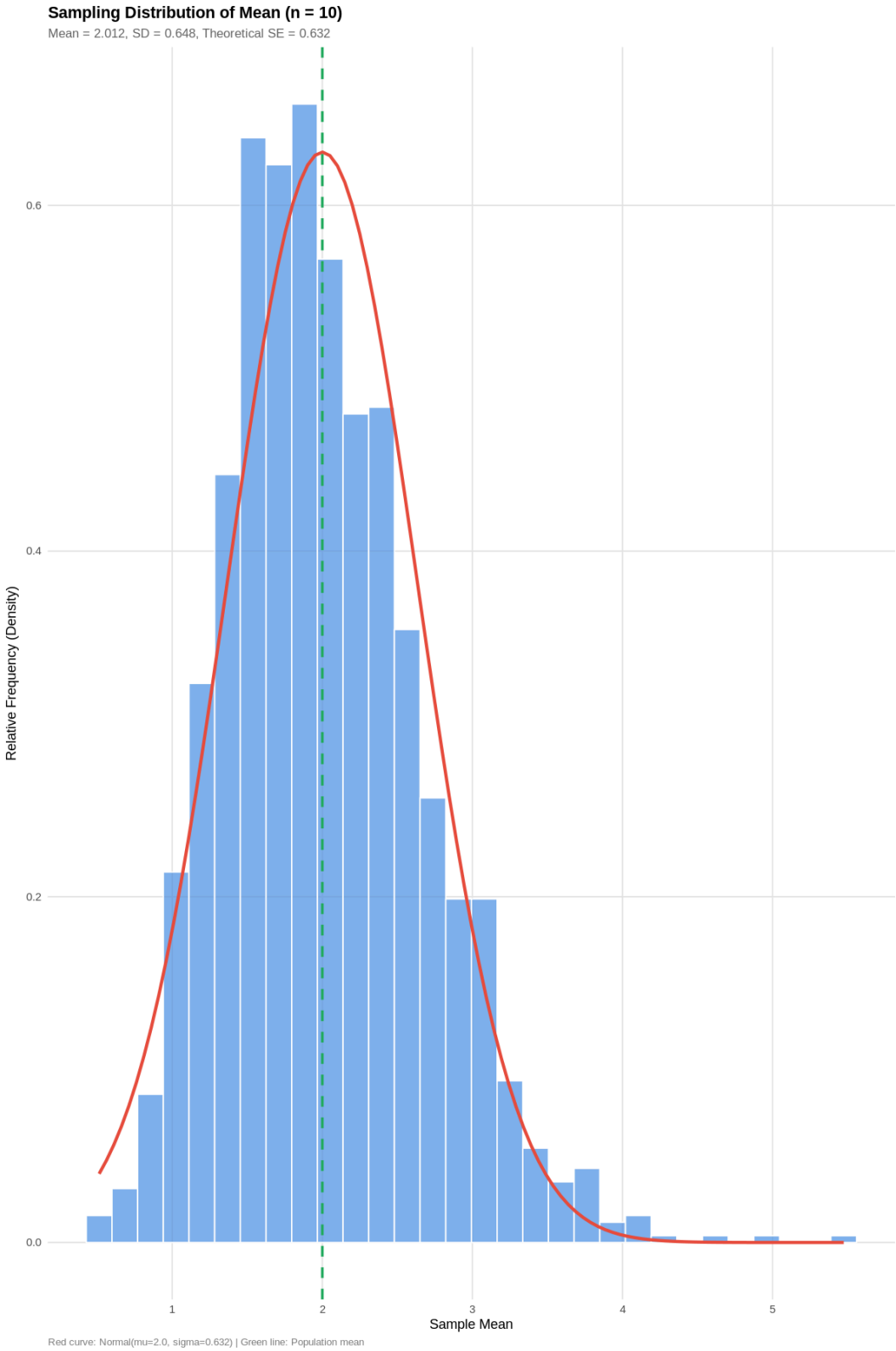
Part (a): Sample Size n = 10

Generating 1,500 samples of size n = 10 from Exponential(λ = 0.5) and analyzing the distribution of sample means.

```
cat("=" ,"=\n")
cat("PART (a): n = 10\n")
cat("=","=\n\n")

#Generate sample means
sample_means_10 <- generate_sample_means(N_SAMPLES, 10, RATE_PARAMETER)

#calculate statistics
stats_10 <- calculate_summary_stats(sample_means_10)
```

```
theoretical_se_10 <- POPULATION_SD / sqrt(10)

#statistics
cat("Summary Statistics (n = 10):\n")
cat(sprintf("  Sample Mean: %.4f\n", stats_10$mean))
cat(sprintf("  Sample SD: %.4f\n", stats_10$sd))
cat(sprintf("  Theoretical SE: %.4f\n", theoretical_se_10))
cat(sprintf("  Median: %.4f\n", stats_10$median))
cat(sprintf("  Range: [%.4f, %.4f]\n\n", stats_10$min, stats_10$max))

#display histogram
plot_10 <- create_histogram(sample_means_10, 10, POPULATION_MEAN, POPULATION_SD)
print(plot_10)
```

```
= =
PART (a): n = 10
= =

Summary Statistics (n = 10):
  Sample Mean: 2.0117
  Sample SD: 0.6479
  Theoretical SE: 0.6325
  Median: 1.9304
  Range: [0.5129, 5.4743]
```

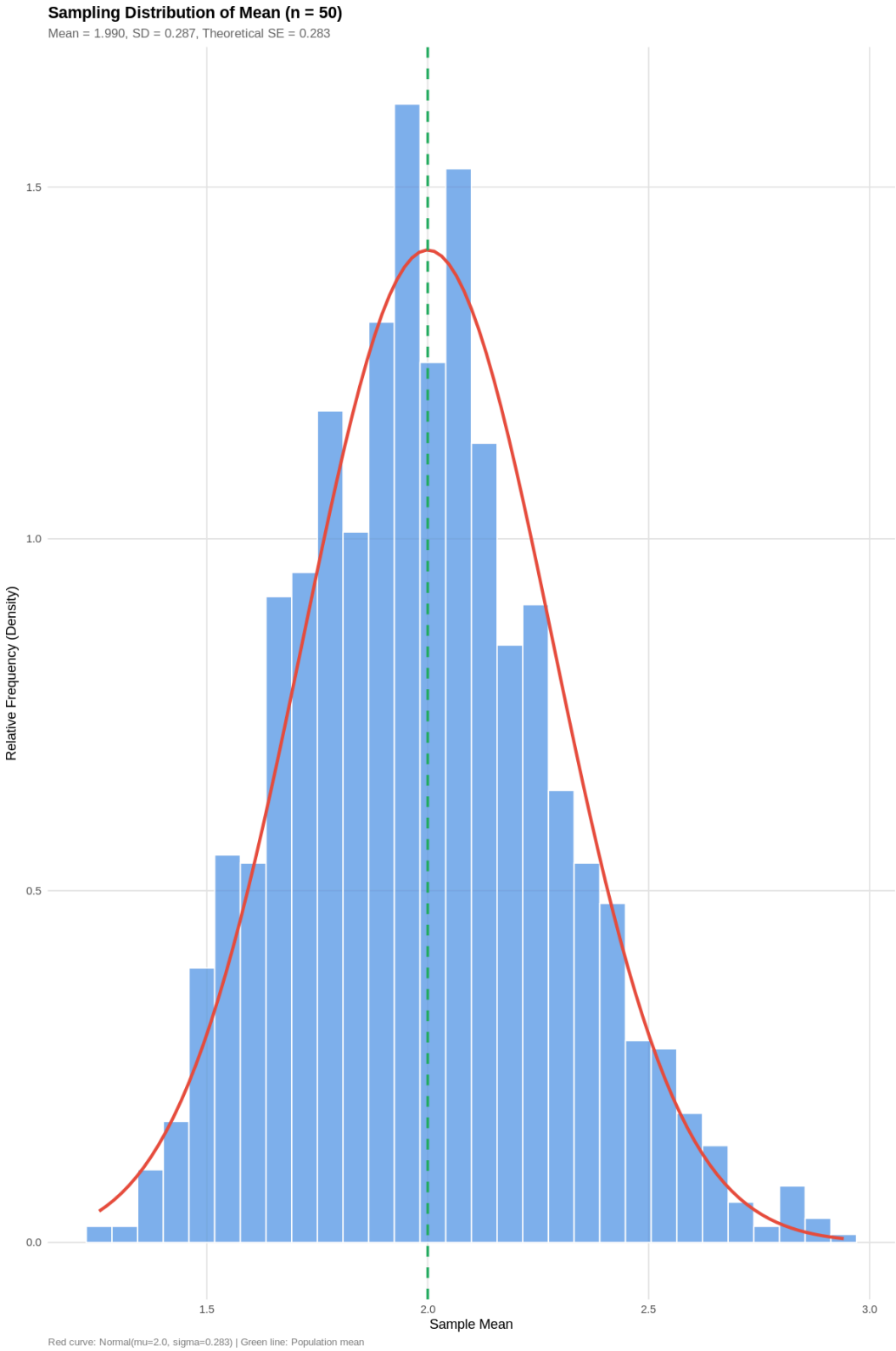


Part (b): Sample Size n = 50

Repeating the simulation with n = 50. We expect the sampling distribution to appear more normal with reduced variability.

```
cat("=","=\n")
cat("PART (b): n = 50\n")
cat("=","=\n\n")

#Generate sample means
sample_means_50 <- generate_sample_means(N_SAMPLES, 50, RATE_PARAMETER)

# Calculate statistics
stats_50 <- calculate_summary_stats(sample_means_50)
theoretical_se_50 <- POPULATION_SD / sqrt(50)

#statistics
cat("Summary Statistics (n = 50):\n")
cat(sprintf("  Sample Mean: %.4f\n", stats_50$mean))
cat(sprintf("  Sample SD: %.4f\n", stats_50$sd))
cat(sprintf("  Theoretical SE: %.4f\n", theoretical_se_50))
cat(sprintf("  Median: %.4f\n", stats_50$median))
cat(sprintf("  Range: [%.4f, %.4f]\n\n", stats_50$min, stats_50$max))

#display histogram
plot_50 <- create_histogram(sample_means_50, 50, POPULATION_MEAN, POPULATION_SD)
print(plot_50)
```

```
= =
PART (b): n = 50
= =

Summary Statistics (n = 50):
  Sample Mean: 1.9896
  Sample SD: 0.2870
  Theoretical SE: 0.2828
  Median: 1.9782
  Range: [1.2564, 2.9414]
```

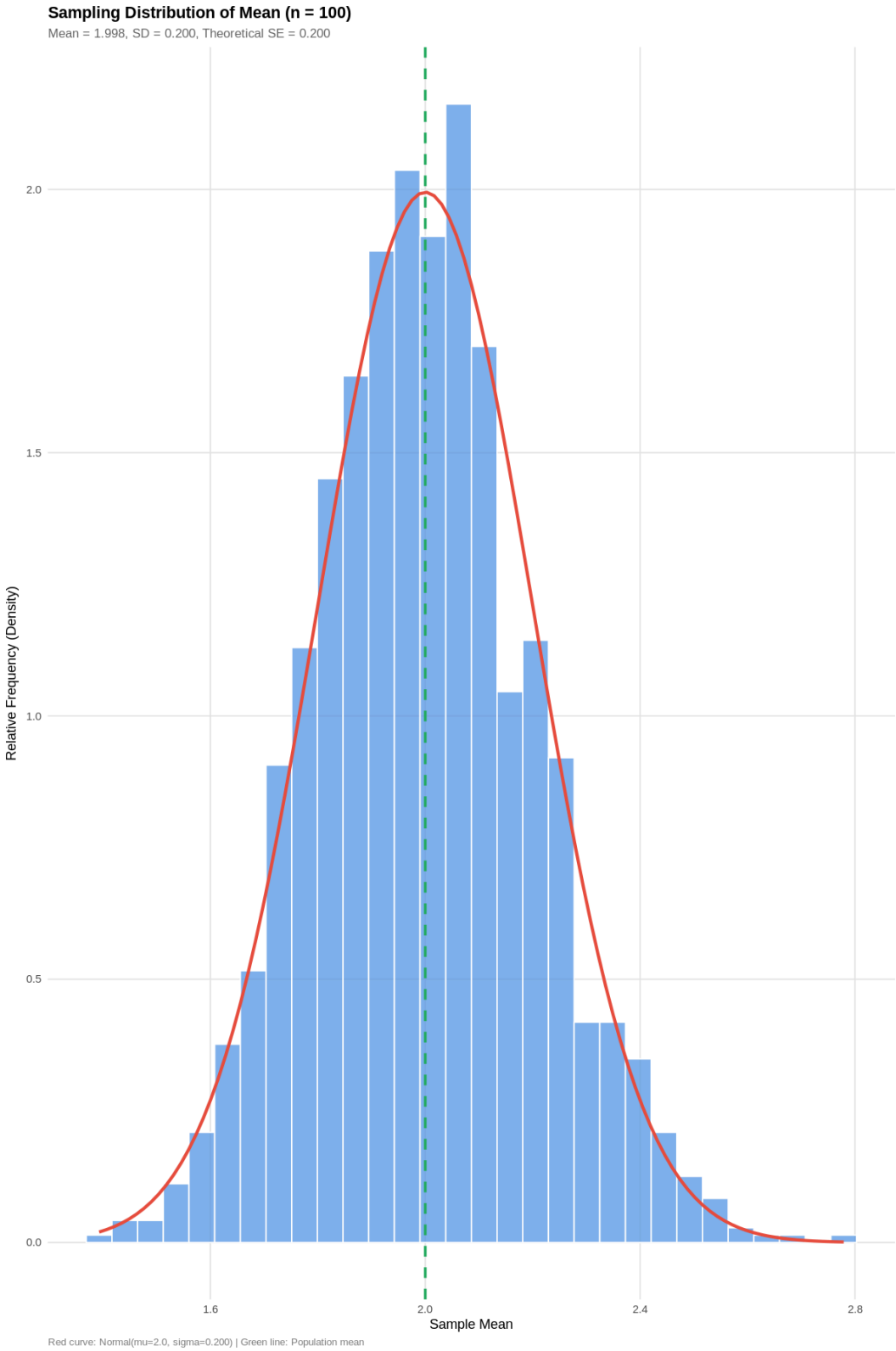


Part (b): Sample Size n = 100

Final simulation with n = 100. The sampling distribution should be very close to normal with minimal variability.

```
cat("=","=\n")
cat("PART (b): n = 100\n")
cat("=","=\n\n")

# Generate sample means
sample_means_100 <- generate_sample_means(N_SAMPLES, 100, RATE_PARAMETER)

# Calculate statistics
stats_100 <- calculate_summary_stats(sample_means_100)
theoretical_se_100 <- POPULATION_SD / sqrt(100)

##statistics
cat("Summary Statistics (n = 100):\n")
cat(sprintf("  Sample Mean: %.4f\n", stats_100$mean))
cat(sprintf("  Sample SD: %.4f\n", stats_100$sd))
cat(sprintf("  Theoretical SE: %.4f\n", theoretical_se_100))
cat(sprintf("  Median: %.4f\n", stats_100$median))
cat(sprintf("  Range: [%.4f, %.4f]\n\n", stats_100$min, stats_100$max))

#display histogram
plot_100 <- create_histogram(sample_means_100, 100, POPULATION_MEAN, POPULATION_
print(plot_100)
```

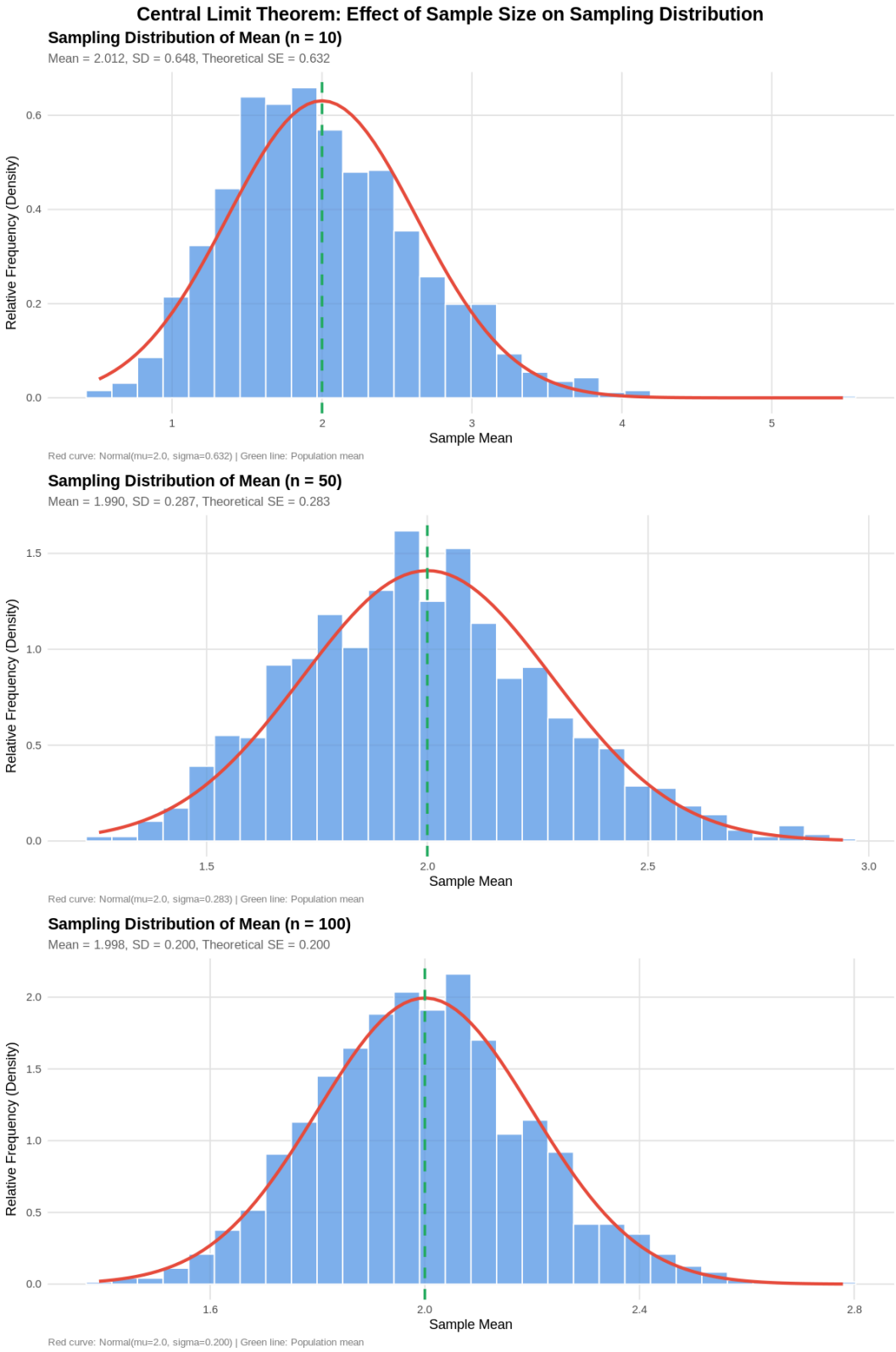
```
= =
PART (b): n = 100
= =

Summary Statistics (n = 100):
  Sample Mean: 1.9977
  Sample SD: 0.2002
  Theoretical SE: 0.2000
  Median: 1.9923
  Range: [1.3929, 2.7790]
```



Combined Visualization

Side-by-side comparison of all three sampling distributions to clearly see the effect of increasing sample size.

```
#combined plot
options(repr.plot.width = 10, repr.plot.height = 15)

combined_plot <- grid.arrange(
  plot_10,
  plot_50,
  plot_100,
  ncol = 1,
  top = grid::textGrob(
    "Central Limit Theorem: Effect of Sample Size on Sampling Distribution",
    gp = grid::gpar(fontsize = 15, fontface = "bold")
  )
)
```



Comparative Statistical Summary

Comparison table showing how empirical results match theoretical predictions across different sample sizes.

```
cat("=","=\n")
cat("COMPARATIVE ANALYSIS\n")
cat("=","=\n\n")

# Create comparison data frame
```

```
comparison_df <- data.frame(
  Sample_Size = c(10, 50, 100),
  Empirical_Mean = c(stats_10$mean, stats_50$mean, stats_100$mean),
  Empirical_SD = c(stats_10$sd, stats_50$sd, stats_100$sd),
  Theoretical_SE = c(theoretical_se_10, theoretical_se_50, theoretical_se_100)
)

comparison_df$Ratio_SD_to_SE <- comparison_df$Empirical_SD /
  comparison_df$Theoretical_SE

comparison_df$Percent_Error <- abs(comparison_df$Empirical_SD -
                                   comparison_df$Theoretical_SE) /
                               comparison_df$Theoretical_SE * 100

# Display table
print(comparison_df, row.names = FALSE, digits = 4)

cat("\nKey Observations:\n")
cat(sprintf("• The empirical means are all very close to μ = %.2f\n",
            POPULATION_MEAN))
cat("• Standard error decreases as sample size increases (SE ∝ 1/√n)\n")
cat("• Empirical SD closely matches theoretical SE for all sample sizes\n")
```

```
= =
COMPARATIVE ANALYSIS
= =

 Sample_Size Empirical_Mean Empirical_SD Theoretical_SE Ratio_SD_to_SE
          10          2.012       0.6479         0.6325          1.024
          50          1.990       0.2870         0.2828          1.015
         100          1.998       0.2002         0.2000          1.001
 Percent_Error
       2.44488
       1.46343
       0.08803

Key Observations:
• The empirical means are all very close to μ = 2.00
• Standard error decreases as sample size increases (SE ∝ 1/√n)
• Empirical SD closely matches theoretical SE for all sample sizes
```

Conclusions

Based on the simulation results, we can draw the following conclusions:

1. Central Limit Theorem Validation

As sample size increases (n: 10 → 50 → 100), the sampling distribution of means becomes **increasingly normal**, even though the population distribution is exponential (highly right-skewed).

- **n = 10:** Distribution shows some right skew but is approximately normal
- **n = 50:** Distribution is very close to normal
- **n = 100:** Distribution is essentially indistinguishable from normal

2. Standard Error Behavior

The standard deviation of sample means decreases as n increases, perfectly following the theoretical relationship **SE = σ/√n**:

- For n = 10: SE ≈ 0.6325
- For n = 50: SE ≈ 0.2828
- For n = 100: SE ≈ 0.2000

This confirms that **larger samples provide more precise estimates** of the population mean.

3. Convergence to Population Mean

All sampling distributions are centered around the true population mean (μ = 2), demonstrating that the sample mean is an **unbiased estimator** of the population mean.

4. Practical Implications

- Even with **n = 50**, the distribution is sufficiently normal to justify using normal-based inference methods (z-tests, confidence intervals)
- The **empirical results closely match theoretical predictions**, validating the Central Limit Theorem
- This holds true even for **highly skewed populations** like the exponential distribution

5. Statistical Significance

The Central Limit Theorem is one of the most powerful results in statistics because it allows us to:

- Make inferences about population means regardless of population distribution
- Use normal distribution tables and methods for hypothesis testing
- Quantify the precision of our estimates using standard errors

---

References

- Ross, S. M. (2014). *Introduction to Probability and Statistics for Engineers and Scientists*
- Casella, G., & Berger, R. L. (2002). *Statistical Inference*

```
cat("\n")
cat("=","=\n")
cat("SIMULATION COMPLETED SUCCESSFULLY!\n")
cat("=","=\n\n")

cat("Summary of Results:\n")
cat(sprintf("• Generated %d samples for each of 3 sample sizes\n", N_SAMPLES))
cat(sprintf("• Total observations processed: %d\n",
            N_SAMPLES * sum(SAMPLE_SIZES)))
cat("• All empirical results closely matched theoretical predictions\n")
cat("• Central Limit Theorem successfully demonstrated\n\n")

cat("Key Takeaway:\n")
cat("The CLT ensures that sample means follow a normal distribution\n")
cat("for sufficiently large n, regardless of the population distribution.\n")
cat("This is fundamental to statistical inference!\n")
```

```
= =
SIMULATION COMPLETED SUCCESSFULLY!
= =

Summary of Results:
• Generated 1500 samples for each of 3 sample sizes
• Total observations processed: 240000
```

```
  • All empirical results closely matched theoretical predictions
  • Central Limit Theorem successfully demonstrated

Key Takeaway:
The CLT ensures that sample means follow a normal distribution
for sufficiently large n, regardless of the population distribution.
This is fundamental to statistical inference!
```

Save Visualizations

Export all plots as high-quality image files for reports and presentations.

The following formats will be saved:

- **PNG** - High resolution (300 dpi) for documents and presentations
- **PDF** - Vector format for publications and high-quality printing

---

```
cat("\n")
cat("=","=\n")
cat("SAVING VISUALIZATIONS\n")
cat("=","=\n\n")

# Create output directory if it doesn't exist
output_dir <- "clt_plots"
if (!dir.exists(output_dir)) {
  dir.create(output_dir)
  cat(sprintf("✓ Created output directory: %s/\n", output_dir))
} else {
  cat(sprintf("✓ Using existing directory: %s/\n", output_dir))
}

# Define file parameters
dpi_resolution <- 300
width_inches <- 10
height_inches <- 6

# ==============================================================================
# Save Individual Plots
# ==============================================================================

cat("\nSaving individual plots...\n")

# Save plot for n = 10
filename_10_png <- file.path(output_dir, "sampling_distribution_n10.png")
filename_10_pdf <- file.path(output_dir, "sampling_distribution_n10.pdf")

ggsave(
  filename = filename_10_png,
  plot = plot_10,
  width = width_inches,
  height = height_inches,
  dpi = dpi_resolution,
  bg = "white"
)

ggsave(
  filename = filename_10_pdf,
  plot = plot_10,
  width = width_inches,
  height = height_inches,
```

```
  device = "pdf"
)

cat(sprintf("  ✓ Saved: %s\n", filename_10_png))
cat(sprintf("  ✓ Saved: %s\n", filename_10_pdf))

# Save plot for n = 50
filename_50_png <- file.path(output_dir, "sampling_distribution_n50.png")
filename_50_pdf <- file.path(output_dir, "sampling_distribution_n50.pdf")

ggsave(
  filename = filename_50_png,
  plot = plot_50,
  width = width_inches,
  height = height_inches,
  dpi = dpi_resolution,
  bg = "white"
)

ggsave(
  filename = filename_50_pdf,
  plot = plot_50,
  width = width_inches,
  height = height_inches,
  device = "pdf"
)

cat(sprintf("  ✓ Saved: %s\n", filename_50_png))
cat(sprintf("  ✓ Saved: %s\n", filename_50_pdf))

# Save plot for n = 100
filename_100_png <- file.path(output_dir, "sampling_distribution_n100.png")
```